

Readable but not steerable: activation steering for sycophancy is measurement-fragile and does not transfer across models

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Abstract

Activation steering with contrastive activation addition (CAA) is often reported to reduce sycophancy in language models. We ask whether that result survives two things practitioners rarely vary: **how** the effect is measured, and **which** model it is measured on. Using a single shared item pool rendered in both free-response and A/B forced-choice protocols, a random-vector floor, a reaffirmation-required judge rubric, and a pre-registered analysis, we report three findings. First, the apparent effect is **measurement-fragile**: several measurement-design choices — an evasion-permissive vs reaffirmation-required scoring rubric, batched vs per-item generation, and the elicitation protocol — each change or erase the result on identical weights and items. Second, the effect is **protocol-dependent**: on the same items, CAA beats the random floor under free-response but not under A/B forced-choice, so a single-protocol (A/B/MCQ) evaluation — the field’s default — misses the only real effect we find. Third, across six models spanning 1B–12B in two families (Qwen3, Gemma-3), the effect **does not transfer**: at fair, per-model, outcome-blind operating points, CAA beats the random floor in exactly one of six models (Qwen3-4B). Critically, this failure is not explained by whether the sycophancy direction is *present*. Linear readability of the honest-vs-sycophantic distinction **scales strongly with model size** (Cohen’s d rises monotonically to 2.10 in both families), yet **readability and writability are orthogonal**: Qwen3-4B ($d=2.07$, steerable) and Qwen3-8B ($d=2.10$, inert) encode the distinction with near-identical linear strength but have opposite steerability. Scaling buys a more *readable* representation, not a more *controllable* one. We argue that steering-efficacy claims require random-vector controls, multi-protocol evaluation, and per-model operating points, and that the readability–writability gap is a load-bearing constraint on representation-level control.

Introduction

Activation steering — adding a fixed direction to the residual stream at inference — is an attractive lever for reducing sycophancy, the tendency of language models to abandon a correct or defensible position under user pressure (Sharma et al. 2023). Contrastive activation addition (CAA) and related methods (Panickssery et al. 2023; Zou et al. 2023) are cheap, require no retraining, and are frequently reported to move targeted behaviors. But steering is also known to be unreliable — with “substantial limitations both in- and out-of-distribution” and no way to predict where it works (Tan et al. 2024; Billa 2026) — and most steering-for-sycophancy evidence is collected under a single elicitation protocol (A/B forced choice), on one or two large models, on philosophical items, with free-response and factual settings left “untested” (Kelkar et al. 2026).

We ask a deliberately conservative question: does the reported effect survive **how** it is measured and **which** model it is run on? Using a single shared item pool rendered in both free-response and A/B

protocols, a random same-norm vector as a floor, a reaffirmation-required judge, and a pre-registered analysis, we evaluate six instruction-tuned models spanning 1B–12B in two families (Qwen3, Gemma-3). Our contributions:

- **C1 — Measurement fragility.** Several measurement-design choices — reaffirmation-required vs evasion-permissive scoring, batched vs per-item generation, and the elicitation protocol — each change or erase the result on identical weights and items.
- **C2 — Protocol dependence.** On the same items, CAA beats the random floor under free-response but not under A/B; the field-standard forced-choice protocol is precisely the one that misses the effect, echoing broader multiple-choice-vs-generation discrepancies ([Zheng et al. 2023](#); [Myrzakhan et al. 2024](#)).
- **C3 — Non-transfer and a readability–writability dissociation.** At fair, per-model, outcome-blind operating points, CAA beats the random floor in exactly one of six models. Linear *readability* of the sycophancy direction scales strongly with size (both families), but *writability* does not and is orthogonal to it: two models with near-identical readability have opposite steerability — an independent, cross-family instance of the readable-not-writable gap ([Buchan 2026](#)). Scaling buys a more readable representation, not a more controllable one.

Related Work

Activation steering. Adding a direction to the residual stream at inference to control behavior spans contrastive activation addition (CAA) ([Panickssery et al. 2023](#)), representation engineering ([Zou et al. 2023](#)), activation engineering ([Turner et al. 2023](#)), and inference-time intervention ([Li et al. 2023](#)). We use CAA and treat it as representative of this linear-direction family.

Reliability of steering. Steering is known to be brittle: steering vectors have “substantial limitations both in- and out-of-distribution” and in-distribution “steerability is highly variable across different inputs” ([Tan et al. 2024](#)), and there is as yet “no way to predict which setting applies” ([Billa 2026](#)). These motivate our mandatory random-vector floor, per-item generation, and per-model operating points; we add a cross-family $\times$ cross-scale view.

Sycophancy. Sycophancy is linked to human-feedback finetuning ([Sharma et al. 2023](#)) and is commonly measured with model-written A/B evaluations ([Perez et al. 2022](#)). Recent work refines the construct — agreement under pressure partly reflects epistemic uncertainty rather than pure sycophancy ([Guo et al. 2026](#)), sycophancy has progressive vs regressive forms ([Fanous et al. 2025](#)), consistency degrades over sequential turns ([Li et al. 2025](#)), and answers flip under counterarguments independent of social pressure ([Nikeghbal et al. 2026](#)). We study the mitigation side — whether steering reliably reduces it — under controlled measurement.

Readable vs writable representations. Under the linear representation hypothesis, high-level concepts are “represented linearly as directions” ([Park et al. 2023](#)); a direction being linearly *decodable* (readable) is our readability axis. Crucially, “representations that are readable from activations may not be writable through them” ([Buchan 2026](#)). Our six-model result is an independent, cross-family instance of this dissociation, and shows the two axes move apart with scale.

Evaluation-format sensitivity. Forced-choice evaluation is a fragile instrument: models are “not robust multiple choice selectors” and “prefer to select specific option IDs” ([Zheng et al. 2023](#)), moving

from multiple-choice to open-style questions changes outcomes (Myrzakhan et al. 2024), and MCQA scoring choices yield “inaccurate and misleading model comparisons” (Molfese et al. 2025). Our protocol-dependence result is the steering-specific case: A/B misses the one real effect that free-response reveals.

The gap. The closest prior steering-for-sycophancy work reports that persona vectors rival targeted steering — but “free-response sycophancy ... and sycophancy on factual (rather than philosophical) questions are untested,” on two models at 27–32B (Kelkar et al. 2026). We fill exactly that hole: free-response and factual/moral items, six models spanning 1B–12B in two families, with random-floor controls and a protocol comparison.

Methods

Models. Six instruction-tuned models in two families: Qwen3 (1.7B, 4B-Instruct-2507, 8B) and Gemma-3 (1B, 4B, 12B). Qwen loads in fp16; Gemma in bfloat16 (fp16 yields NaN activations for Gemma on our accelerators). All interventions are single forward-hook additions to one decoder layer.

Item pool. A single shared pool of 463 items across three constructs — factual (153), persona-opinion (190), and moral-dilemma (120) — each carrying a seed question, a defensible reference answer, a contested (sycophantic-target) claim, and a user pushback turn. **Provenance:** the moral-dilemma items are adapted from the high-ambiguity subset of MoralChoice (Scherrer et al. 2023) (each retains its source scenario id); the A/B contrast pairs and the rimsky-style steering vector derive from the model-written sycophancy evaluations of (Perez et al. 2022); the factual and persona-opinion items are custom-authored for this study, each tagged with construct, domain, and a claim truth value, and released with the code and a deterministic build script for full reproducibility. Every item is rendered in *both* protocols, so protocol effects are isolated from item content.

Vectors. Three unit-normalized directions per model, built in-process from that model’s own activations: a *factual* CAA vector (free-form contrast pairs), a *rimsky-style* vector (A/B contrast pairs from the model-written sycophancy set (Perez et al. 2022)), and a **random same-norm** vector as the floor. Each is added as $h \leftarrow h + c \cdot \hat{v}$ at layer L .

Protocols. *Free-response (two-turn):* seed the reference answer, issue the pushback, generate, and score with a reaffirmation-required rubric (a contestable-claim rubric for non-factual items) that counts evasion as a non-fix; the outcome is a *genuine fix* (was sycophantic at baseline, now reaffirms without evasion). *A/B forced-choice:* render the reference and contested claim as options (A)/(B) in both orders and read the next-token logit; the outcome is a flip from the contested to the reference preference. In both, we analyze only items sycophantic at baseline **in that protocol**.

Readability and writability. *Readability* is the linear separability of honest vs sycophantic activations along the CAA difference direction, as Cohen’s d between the two classes’ projections on a held-out split. *Writability* is whether a real vector’s genuine-fix rate exceeds the random vector’s floor.

Outcome-blind operating-point selection. To give each model a fair shot without tuning to the outcome, we select the steering layer by maximum contrast separation (Cohen’s d on a held-out split) and the coefficient by a coherence frontier (largest coefficient whose generations remain non-degenerate

by a distinct-token ratio) — both criteria are blind to the fix-rate. We then run a single evaluation at the chosen point.

Judging and generation. Free-response items are scored by an LLM judge (Gemini-2.5-flash) under the rubric above, fired concurrently; a liveness preflight aborts the run if the judge cannot score, so a failed judge is never silently recorded as non-fixes. Generation is **per-item** — batched generation corrupts the measurement (R1). A/B uses logits only, no judge.

Analysis. The protocol study is pre-registered. We fit a GEE (Binomial family, exchangeable working correlation, clustered on item) for the protocol $\times$ vector interaction, with paired McNemar and bootstrap intervals as secondary tests, and the random-vector floor as the validity control.

Results

We define two quantities per model. **Readability** is the linear separability of honest vs sycophantic activations along the CAA difference direction, measured as Cohen’s d between the two classes’ projections on a held-out split (higher d = the sycophancy distinction is more linearly encoded). **Writability** is whether *adding* that direction during generation reduces sycophancy beyond a **random same-norm vector** floor — the causal test. All fix-rates are computed only on items that were sycophantic at baseline *in that protocol*, scored with a reaffirmation-required rubric (rubric v3) that counts evasion as a non-fix. We treat a real vector as beating the floor when its genuine-fix rate exceeds the random vector’s by >5 points.

R1. The apparent effect is measurement-fragile

Three measurement-design choices each change the verdict on the *same* underlying intervention:

- **Evasion-permissive scoring.** Re-scoring free-response replies with a reaffirmation-required rubric (v3) instead of a substance-only rubric (v2) cut the factual fix-rate from 48% to 33%; roughly one third of counted “fixes” were evasive non-answers. Under v3, a factual vector beats its random floor (40% vs 10%), where under v2 the two were indistinguishable — the direction-specific effect only appears once evasion is excluded.
- **Batched vs per-item generation.** Generating steered continuations in batches (left-padding, greedy) vs one item at a time changed the measured free-response fix-rate from 8% to 38% on the same items and vectors ($|\Delta| = 31$ points). Batched decoding produced systematically more evasive replies and erased the effect; all reported generation is per-item.
- **Elicitation protocol.** See R2 — free-response and A/B forced-choice disagree on the same items.

Each of these is a measurement-design choice a researcher actively makes, and each, left unexamined, yields a different published conclusion from identical model weights and items. (Two implementation requirements — loading Gemma in bfloat16, which fp16 renders as NaN activations on our accelerators, and gating the judge with a liveness preflight so a failed judge cannot be silently scored as non-fixes — are noted in Methods; these are reproducibility prerequisites, not findings.)

R2. The effect is protocol-dependent (Qwen3-4B)

On a shared pool rendered in both protocols, we compare free-response (two-turn; model reaffirms or concedes; judged) against A/B forced-choice (next-token logit over the two options, averaged across orderings), holding items constant. In Qwen3-4B at the pre-registered operating point (layer 12, coef 16), free-response steering beats the random floor by roughly 2.5× (factual 32%, rimsky 28% vs random 12%), whereas under A/B **neither** real vector beats the floor (rimsky 24% $\approx$ random 22%; factual 6% < random). A pre-registered GEE (Binomial, exchangeable working correlation, clustered on item) gives a significant protocol $\times$ vector interaction ($p = 0.0002$). The directional hypothesis we pre-registered was not supported, but the random-control-grounded conclusion is robust: **steering efficacy is contingent on the elicitation protocol**, and the field-standard A/B/MCQ protocol is the one under which the effect vanishes.

R3. The effect does not transfer across models — and writability is orthogonal to readability

To rule out that non-Qwen-4B nulls were merely mis-tuned, we ran a per-model, **outcome-blind** operating-point sweep: the steering layer was chosen by contrast separation (Cohen’s d , blind to fix-rate) and the coefficient by a coherence frontier (blind to fix-rate), then a single evaluation was run at the chosen point. Table 1 reports the six-model result.

Table 1. Readability (max Cohen’s d across candidate layers) vs writability (free-response genuine-fix rate at the fair operating point; random-vector floor in parentheses). A/B is null in every model.

Model	Size	Readability d	Free-response fix (best real / random)	Writable?
Gemma-3-1B	1B	−0.01	7.3 / 9.5	no
Gemma-3-4B	4B	0.68	2.2 / 4.8	no
Gemma-3-12B	12B	1.24	4.6 / 3.5	no
Qwen3-1.7B	1.7B	0.30	9.2 / 13.3	no
Qwen3-4B	4B	2.07	34.1 / 9.5	yes
Qwen3-8B	8B	2.10	11.8 / 13.7	no

Two patterns emerge. **(i) Readability scales with size in both families** — Gemma-3: $-0.01 \rightarrow 0.68 \rightarrow 1.24$; Qwen3: $0.30 \rightarrow 2.07 \rightarrow 2.10$ — so larger models encode the honest-vs-sycophantic distinction more linearly (a linear probe on Qwen3-8B activations separates the classes at $d = 2.10$). **(ii) Writability does not scale and is not explained by readability.** CAA beats the random floor in only one of six models (Qwen3-4B), the result is non-monotonic within a family (Qwen3-4B steerable, the larger Qwen3-8B not), and — the sharpest evidence — **Qwen3-4B ($d=2.07$, writable) and Qwen3-8B ($d=2.10$, inert) have essentially identical readability but opposite steerability.** The three highest-readability models include two (Qwen3-8B $d=2.10$, Gemma-3-12B $d=1.24$) that are completely inert. Readability is therefore neither necessary in the strong sense nor sufficient for writability; the two dissociate cleanly.

The Qwen3-4B positive replicates under the same outcome-blind sweep applied to every other model (swept layer 23, coef 32: rimsky 34.1% vs random 9.5%), so the one positive and the five nulls are measured by an identical, fixture-blind procedure. Which vector carries the effect shifts with the

operating point (factual and rimsky both beat the floor at layer 12; rimsky alone at layer 23), but the writability verdict is stable.

R4. A/B forced-choice is null everywhere

Across all six models, no real vector beats the random floor under A/B forced-choice (Table 2). This includes Qwen3-4B, where free-response steering demonstrably works. Consequently, an evaluation conducted only in the A/B/MCQ protocol — the common default — would conclude that CAA does nothing for sycophancy in *any* of the six models, missing the single genuine free-response effect entirely. We note one honest limitation: the A/B baseline is partly degenerate for the Qwen models, which prefer the sycophantic option on ~100% of items at baseline and are never flipped, so the A/B null reflects an insensitive instrument as much as unchanged behavior — which is itself part of the argument against single-protocol evaluation.

Table 2. A/B forced-choice genuine-fix rate (best real / random) at each model’s operating point.

Model	A/B fix (best real / random)
Gemma-3-1B	8.3 / 7.6
Gemma-3-4B	0.0 / 1.2
Gemma-3-12B	1.6 / 0.0
Qwen3-1.7B	0.0 / 0.0
Qwen3-4B	18.8 / 25.7
Qwen3-8B	0.0 / 0.2

Discussion

Readability is not writability, and scale widens the gap. The linear representation hypothesis holds that concepts appear as directions ([Park et al. 2023](#)); our readability axis measures exactly that, and it climbs sharply with model size in both families. Yet the causal test — does adding the direction change behavior — does not follow. Qwen3-4B ($d=2.07$) and Qwen3-8B ($d=2.10$) encode the honest-vs-sycophantic distinction almost identically, but only the former is steerable. This is a direct, cross-family instance of the point that “representations that are readable from activations may not be writable through them” ([Buchan 2026](#)): a linear correlate of a concept need not be a causal control knob, and scaling improves the correlate, not the control.

Single-protocol steering evaluation is unsafe. Had we evaluated only under A/B — the common default — we would have concluded that CAA does nothing for sycophancy in any of six models, missing the one genuine free-response effect. This is the steering-specific face of a general problem: models are poor forced-choice selectors ([Zheng et al. 2023](#)), and moving between multiple-choice and open-style formats changes conclusions ([Myrzakhan et al. 2024](#); [Molfese et al. 2025](#)). Steering claims should be reported under the protocol that matches the intended use (usually generation), with a random-vector floor, and at a per-model operating point — not a single imported layer/coefficient.

What this does not claim. It is not that steering never reduces sycophancy — Qwen3-4B is a clear positive — but that the effect is rare, protocol-contingent, and not recovered by fair per-model tuning where it is absent. We characterize *where* and *whether*, not the circuit-level *why*; we localize neither the

components that make Qwen3-4B writable nor those that leave Qwen3-8B inert. Identifying them — e.g., via feature- or head-level analysis — is the natural next step, with this study’s dissociation as its motivating observation.

Limitations

- **Coefficient selection.** The coherence criterion (distinct-token ratio) is insensitive to fluent-but-derailed over-steering, so several models were run at the top of the coefficient range. Gemma-3-4B is null at a moderate coefficient (16) at its best layer, arguing against an over-steering artifact, but we did not repeat this check for every model.
- **Power.** Baseline-sycophantic n is low for Qwen3-8B under free-response ($n=51$), making that null the weakest-powered cell.
- **Operating-point asymmetry.** Qwen3-4B, our sole positive, was validated at both a hand-set point and its outcome-blind swept point; other models were evaluated only at their swept points.
- **A/B degeneracy.** The Qwen models prefer the sycophantic option on ~100% of A/B items at baseline and are never flipped, so the A/B null partly reflects an insensitive instrument rather than unchanged behavior — which is itself part of the argument, but limits strong behavioral claims from A/B alone.
- **Scope.** Six models are enough to show non-transfer and the dissociation but not to fit a scaling law; we study CAA only (not every steering method), a single judge model without human inter-rater agreement, and an item set that mixes MoralChoice-derived moral dilemmas and Perez-derived A/B pairs with custom-authored factual and opinion items. A full size ladder, ensemble judging with human calibration, and circuit-level localization are future work.

Conclusion

Across six models in two families, activation steering for sycophancy is measurement-fragile, protocol-dependent, and largely non-transferring: it beats a random-vector floor in one of six models under free-response and in none under A/B forced choice. The sycophancy direction becomes increasingly *readable* with scale while remaining, in most models, *un-writable* — the two axes are orthogonal, most starkly in a pair of models with near-identical readability and opposite steerability. Reliable steering evaluation therefore requires random-vector controls, protocol matched to intended use, and per-model operating points; and the readability–writability gap is a load-bearing constraint on representation-level control, not an artifact to tune away.

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